Nonparametric kernel regression with multiple predictors and multiple shape constraints

Pang Du[1] Christopher F. Parmeter[2] Jeffrey S. Racine[3]

[1]Virginia Tech
Blacksburg, VA USA 24061-0002

[2]University of Miami
Coral Gables, FL USA 33124-6520

[3]McMaster University
Hamilton, ON Canada L8S 4M4

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Summary

- Develop a nonparametric estimator capable of handling general derivative constraints.
- Develop hypothesis test on constraints.
- Investigate the asymptotic properties of the estimation and test procedures.
- Simulated and empirical evidence of the technique.

Overview

- Economic theory often imposes constraints upon underlying relationships (i.e., monotonicity, concavity, homogeneity, etc.).
- No general approach exists for constructing constrained smooth nonparametric kernel regression, which severely limits its applicability.
- Powerful methods have been proposed for imposing monotonicity in nonparametric settings; see Hall, Huang, Gifford, & Gijbels (2001) and Hall & Huang (2001).
- We generalize Hall & Huang's (2001) approach to allow for equality or inequality constraints on a nonparametric regression model and its derivatives of any order.
- A testing framework is provided allowing researchers to thereby impose and test the validity of the restrictions.
- Illustrative Monte Carlo results are presented, and an application is considered.

Existing Approaches

- Imposing constraints on a nonparametric estimate can potentially produce *statistically consistent* estimators that are *(econ-)theoretically consistent*.
- We also may wish to test the validity of the constraints.
- The existing literature falls into four broad areas:
  - nonparametric estimators for a particular constraint (e.g., monotonicity).
  - nonparametric estimators for a class of constraints (e.g., constrained smoothing splines).
  - tests for a particular constraint (e.g., monotonicity).
  - tests for a class of constraints (e.g., conditional moment test for constrained splines).

Previous Work: Constrained Flexible Estimation

- Gallant (1981) introduced the Fourier Flexible Form estimator (FFF) which is a series-based estimator that can be restricted (though monotonicity is not easily imposed in this setting).
- Matzkin (1991) considered identification and estimation of general nonparametric problems with economic constraints (when identification is not possible, imposing shape constraints tied to economic theory may provide nonparametric identification).
- Yatchew & Bos (1997) develop a series-based estimator that can handle general constraints.
- Beresteanu (2004) introduced a spline-based procedure that can handle multiple, general, derivative constraints.

But, there does not yet exist a methodology grounded in kernel methods that can impose general constraints and statistically test the validity of these constraints.

Proposed Approach

- $\{\mathbf{X}_i, Y_i\}_{i=1}^n$: sample pairs of explanatory and response variables.
- Our goal is to estimate the unknown mean response $g(\mathbf{x}) = E(Y|\mathbf{X} = \mathbf{x})$ subject to constraints on $g^{(\mathbf{s})}(\mathbf{x})$ where $\mathbf{s}$ is a k-vector corresponding to the dimension of $\mathbf{x}$.
- The elements of $\mathbf{s}$ represent the order of the partial derivative corresponding to each element of $\mathbf{x}$.
- Thus $\mathbf{s} = (0, 0, \ldots, 0)$ represents the function itself, while $\mathbf{s} = (1, 0, \ldots, 0)$ represents $\partial g(\mathbf{x})/\partial x_1$.
- In general, for $\mathbf{s} = (s_1, s_2, \ldots, s_k)$ we have

$$g^{(\mathbf{s})}(\mathbf{x}) = \frac{\partial^{s_1} g(\mathbf{x})}{\partial x_1^{s_1}} \cdots \frac{\partial^{s_k} g(\mathbf{x})}{\partial x_k^{s_k}} \tag{1}$$

- We consider the class of kernel regression smoothers that can be written as linear combinations of the response Y_i, i.e.,

$$\hat{g}(\mathbf{x}) = \sum_{i=1}^{n} A_i(\mathbf{x}) Y_i, \tag{2}$$

  where $A_i(\cdot)$ is a local weight function.
- We presume that the reader may wish to impose constraints on the estimate $\hat{g}(\mathbf{x})$ of the form $l(\mathbf{x}) \leq \hat{g}^{(\mathbf{s})}(\mathbf{x}) \leq u(\mathbf{x})$ for arbitrary $u(\cdot)$, $l(\cdot)$, and $\mathbf{s}$

- Following Hall & Huang (2001), we consider a generalization of $\hat{g}(\mathbf{x})$ defined in (2) given by

$$\hat{g}(\mathbf{x}|p) = \sum_{i=1}^{n} p_i A_i(\mathbf{x}) Y_i \tag{3}$$

- For what follows $\hat{g}^{(\mathbf{s})}(\mathbf{x}|p) = \sum_{i=1}^{n} p_i A_i^{(\mathbf{s})}(\mathbf{x}) Y_i$ where $A_i^{(\mathbf{s})}(\mathbf{x}) = \frac{\partial^{s_1} A_i(\mathbf{x})}{\partial x_1^{s_1}} \cdots \frac{\partial^{s_k} A_i(\mathbf{x})}{\partial x_k^{s_k}}$ for continuous $\mathbf{x}$.
- Again, in our notation $\mathbf{s}$ represents a $k \times 1$ vector of nonnegative integers that indicate the order of the partial derivative of the weighting function of the kernel smoother.

- By way of example, using (3) to generate an unrestricted local constant estimator (Nadaraya-Watson estimator) we would set $p_i = 1/n$, $i = 1, \ldots, n$, and set

$$A_i(\mathbf{x}) = \frac{nK_\gamma(\mathbf{X}_i, \mathbf{x})}{\sum_{j=1}^n K_\gamma(\mathbf{X}_j, \mathbf{x})}, \tag{4}$$

  where $K_\gamma(\cdot)$ is a generalized product kernel that admits both continuous and categorical data, and γ is a vector of bandwidths.
- When $p_i \neq 1/n$ for some i, then we would have a constrained local constant estimator.
- Note that one uses the same bandwidths for the constrained and unconstrained estimator hence bandwidth selection proceeds using standard methods.

Quick Digression on Kernel Estimation

- The local constant kernel estimator of the unknown conditional mean can be expressed as

$$\hat{g}(\mathbf{x}) = (\mathbf{i}'\mathcal{K}(\mathbf{x})\mathbf{i})^{-1}\mathbf{i}'\mathcal{K}(\mathbf{x})\mathcal{Y}, \tag{5}$$

  where $\mathbf{i}$ is an n-vector of ones, $\mathcal{Y}$ is our vector of responses and $\mathcal{K}(\mathbf{x})$ is a diagonal matrix with $A_i(\mathbf{x})$ (as in (4)) as the ith element.
- If we only had an intercept then our least squares regression would be

$$\hat{g}(\mathbf{x}) = (\mathbf{i}'\mathbf{i})^{-1}\mathbf{i}'\mathcal{Y}$$

  which looks almost identical to our local constant estimator, aside from the kernel matrix $\mathcal{K}(\mathbf{x})$.

- Let p_u be an n-vector with elements $1/n$ and let p be the vector of weights to be selected.
- In order to impose our constraints, we choose $p = \hat{p}$ to minimize the distance from p to the uniform weights $p_i = 1/n$ $\forall i$.
- This is appealing intuitively since the unconstrained estimator is that for which $p_i = 1/n$ $\forall i$, as noted above.
- Whereas Hall & Huang (2001) consider probability weights (i.e., $0 \leq p_i \leq 1$, $\sum_i p_i = 1$) and distance measures suitable for probability weights (i.e., Hellinger), we shall need to relax the constraint that $0 \leq p_i \leq 1$ and will instead allow for both positive and negative weights (while retaining $\sum_i p_i = 1$), and shall also therefore require alternative distance measures.

- We shall forgo the power divergence metric of Cressie & Read (1984) which was used by Hall & Huang (2001) since it is only valid for probability weights
- For what follows we select the well-worn L_2 metric $D(p) = (p_u - p)'(p_u - p)$ which has a number of appealing features in this context
- Our problem therefore boils down to selecting those weights p that minimize $D(p)$ subject to $l(\mathbf{x}) \leq \hat{g}^{(\mathbf{s})}(\mathbf{x}) \leq u(\mathbf{x})$.
- This can be cast as a general nonlinear programming problem which can be solved using standard quadratic programming methods and off-the-shelf software.

- The two-sided constraint $l(\mathbf{x}) \leq \hat{g}^{(\mathbf{s})}(\mathbf{x}) \leq u(\mathbf{x})$ can be considered as a special case of multiple simultaneous one-sided constraints:

$$\sum_{\mathbf{s} \in \mathbf{S}_k} \alpha_{\mathbf{s},k} \hat{g}^{(\mathbf{s})}(\mathbf{x}) - c_k(\mathbf{x}) \geq 0, k = 1, \ldots, T,$$

  where T is the number of restrictions, and in each restriction the sum is taken over all vectors in $\mathbf{S}_k$ that correspond to our constraints and $\alpha_{\mathbf{s},k}$ is a set of constants used to generate various constraints.
- In what follows we shall presume, without loss of generality, that for all $\mathbf{s}$, $\alpha_{\mathbf{s},k} \geq 0$ and $c_k(\mathbf{x}) \equiv 0$, since the $c_k(\mathbf{x})$'s are known functions.
- Differential operator notation $m \mapsto m^{\mathcal{D}}$ such that $m^{\mathcal{D}}(\mathbf{x}) = \sum_{\mathbf{s} \in \mathbf{S}_k} \alpha_{\mathbf{s},k} m^{(\mathbf{s})}(\mathbf{x})$.

Existence and Uniqueness

Theorem (Theorem 3.1)

Assume that the set $\{1, \ldots, n\}$ *contains a sequence* $\{i_1, \ldots, i_k\}$ *with the following properties.*

(i) *For each* k, $A^{\mathcal{D}}_{i_k}(\mathbf{x})$ *is strictly positive and continuous on an open set* $\mathbf{O}_{i_k} \subset \mathbb{R}^r$, *and vanishes on* $\mathbb{R}^r \setminus \mathbf{O}_{i_k}$,

(ii) *Every* $\mathbf{x} \in \mathcal{J} = \prod_{i=1}^{r} [a_i, b_i]$ *is contained in at least one open set* $\mathbf{O}_{i_k}$,

(iii) *For* $1 \leq i \leq n$, $A^{\mathcal{D}}_{i_k}(\mathbf{x})$ *is continuous on* $(-\infty, \infty)^r$,

then there exists a vector $p = (p_1, \ldots, p_n)$ *such that the constraints are satisfied for all* $x \in \mathcal{J}$.

Assumption (Assumption 3.1)

(i) The sample $\mathbf{X}_i$ either form a regularly spaced grid on a compact set $\mathcal{I} \equiv \prod_{i=1}^{r}[c_i, e_i]$ or constitute independent random draws from a distribution whose density f is continuous and nonvanishing on $\mathcal{I}$; the ε_i are independent and identically distributed with zero mean and variance σ^2, and are independent of the $\mathbf{X}_i$; the kernel function $K(\cdot)$ is a symmetric, compactly supported density such that $K^{\mathcal{D}}$ is Hölder-continuous on $\mathcal{J} \subset \mathcal{I}$.

(ii) $E\left(|\varepsilon_i|^t\right)$ is bounded for sufficiently large $t > 0$.

(iii) $g^{\mathcal{D}}$ is continuous on $\mathcal{J}$. For random $\mathbf{X}_i$'s, their density function f is also continuous on $\mathcal{J}$.

(iv) The bandwidth associated with each explanatory variable, h_j, satisfies $h_j \propto n^{-1/(3r+2|\mathbf{d}|)}$, $1 \leq j \leq r$, where $|\mathbf{d}| = \sum_{j=1}^{r} d_j$ is the maximum order of the derivative vector $\mathbf{d}$ in $\mathcal{D}$.

Theorem (Theorem 3.2)

Assume 3.1 (i)-(iv) hold

(i) *If* $g^{\mathcal{D}} > 0$ *on* $\mathcal{J}$ *then, with probability 1,* $\hat{p} = 1/n$ *for all sufficiently large* n *and* $\hat{g}^{\mathcal{D}}(\cdot|\hat{p}) = \tilde{g}^{\mathcal{D}}$ *on* $\mathcal{J}$ *for all sufficiently large* n*. Hence,* $\hat{g}(\cdot|\hat{p}) = \tilde{g}$ *on* $\mathcal{J}$ *for all sufficiently large* n*.*

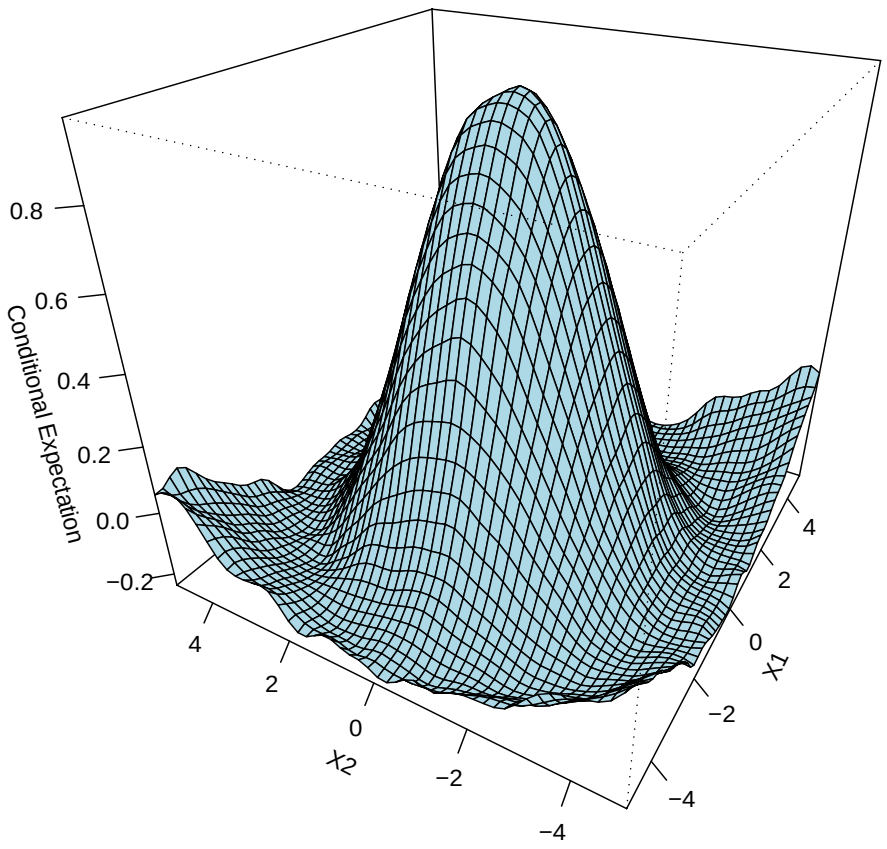
(ii) *Suppose that* $g^{\mathcal{D}} > 0$ *except on an interior hyperplane subset* $\mathcal{X}_0 \subset \mathcal{J}$ *such that* $g^{\mathcal{D}}(\mathbf{x}) = 0, \forall \mathbf{x} \in \mathcal{X}_0$*. Also, for any* $\mathbf{x}_0 \in \mathcal{X}_0$*, suppose that* $g^{\mathcal{D}}$ *has second order continuous derivatives in the neighborhood of* $\mathbf{x}_0$ *with* $\frac{\partial g^{\mathcal{D}}}{\partial \mathbf{x}}(\mathbf{x}_0) = \mathbf{0}$ *and with* $\frac{\partial^2 g^{\mathcal{D}}}{\partial \mathbf{x} \partial \mathbf{x}^T}(\mathbf{x}_0)$ *nonsingular; then* $|\hat{g}(\cdot|\hat{p}) - \tilde{g}| = O_p\left(h^{|\mathbf{d}| + \frac{r+1}{2}}\right)$ *uniformly on* $\mathcal{J}$*.*

Theorem (Theorem 3.2)

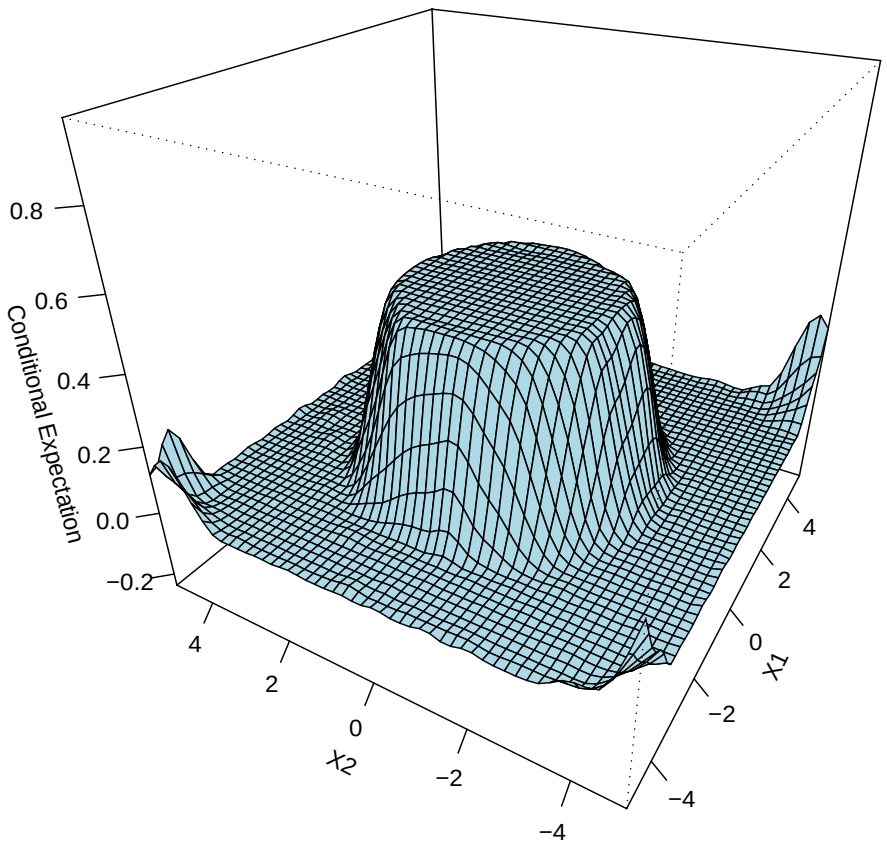
(iii) *Under the same conditions given in (ii) above, there exist random variables* $\Theta = \Theta(n)$ *and* $Z_1 = Z_1(n) \geq 0$ *satisfying* $\Theta = O_p\left(h^{|\mathbf{d}|+r+1}\right)$ *and* $Z_1 = O_p(1)$ *such that* $\hat{g}(\mathbf{x}|\hat{p}) = (1+\Theta)\tilde{g}(\mathbf{x})$ *uniformly for* $\mathbf{x} \in \mathcal{J}$ *with* $\inf_{\mathbf{x}_0 \in \mathcal{X}_0} |\mathbf{x} - \mathbf{x}_0| > Z_1 h^{(r+1)/4}$.

The latter property reflects the fact that, for a random variable $Z_2 = Z_2(n) \geq 0$ *satisfying* $Z_2 = O_p(1)$, *we have* $\hat{p}_i = n^{-1}(1+\Theta)$ *for all indices* i *such that both* $\inf_{\mathbf{x}_0 \in \mathcal{X}_0} |X_i - \mathbf{x}_0| > Z_2 h^{(r+1)/4}$ *and* $A_i(\mathbf{x}) \neq 0$ *for some* $\mathbf{x} \in \mathcal{J}$.

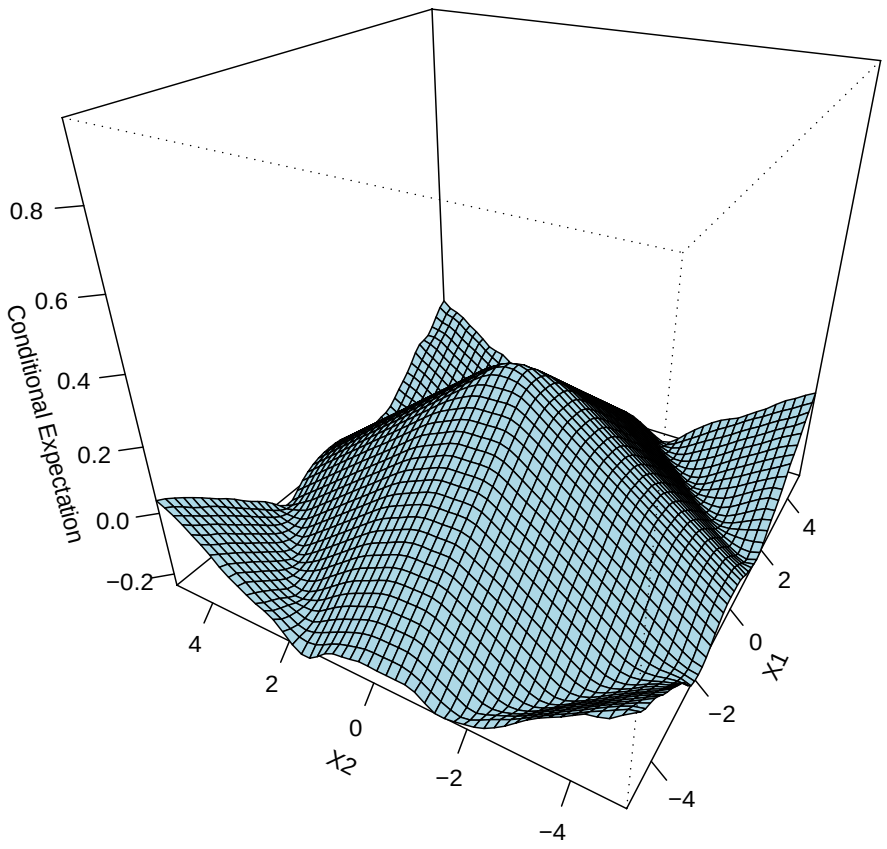
$g(\mathbf{x}) = \sin\left(\sqrt{x_1^2 + x_2^2}\right) / \sqrt{x_1^2 + x_2^2}$



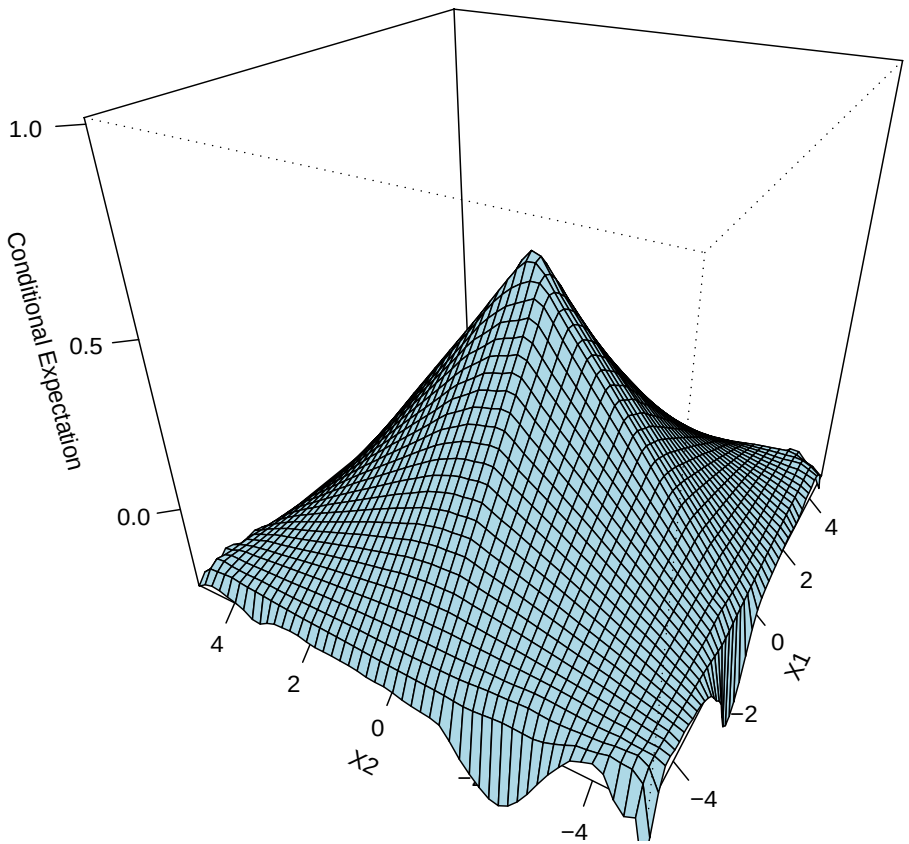
$g(\mathbf{x}) = \sin\left(\sqrt{x_1^2 + x_2^2}\right) / \sqrt{x_1^2 + x_2^2}$, $0 \leq \hat{g}(\mathbf{x}|p) \leq 0.5$



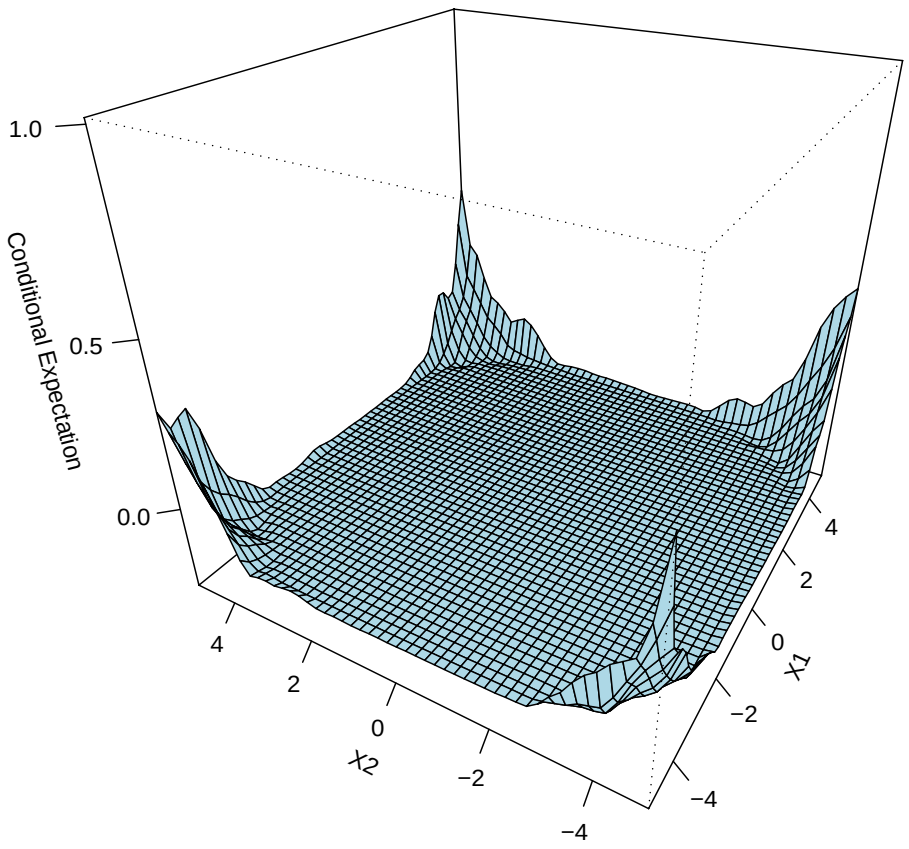
$g(\mathbf{x}) = \sin\left(\sqrt{x_1^2 + x_2^2}\right) / \sqrt{x_1^2 + x_2^2}$, $-0.1 \leq \partial\hat{g}(\mathbf{x}|p)/\partial x_1 \leq 0.1$, $-0.1 \leq \partial\hat{g}(\mathbf{x}|p)/\partial x_2 \leq 0.1$



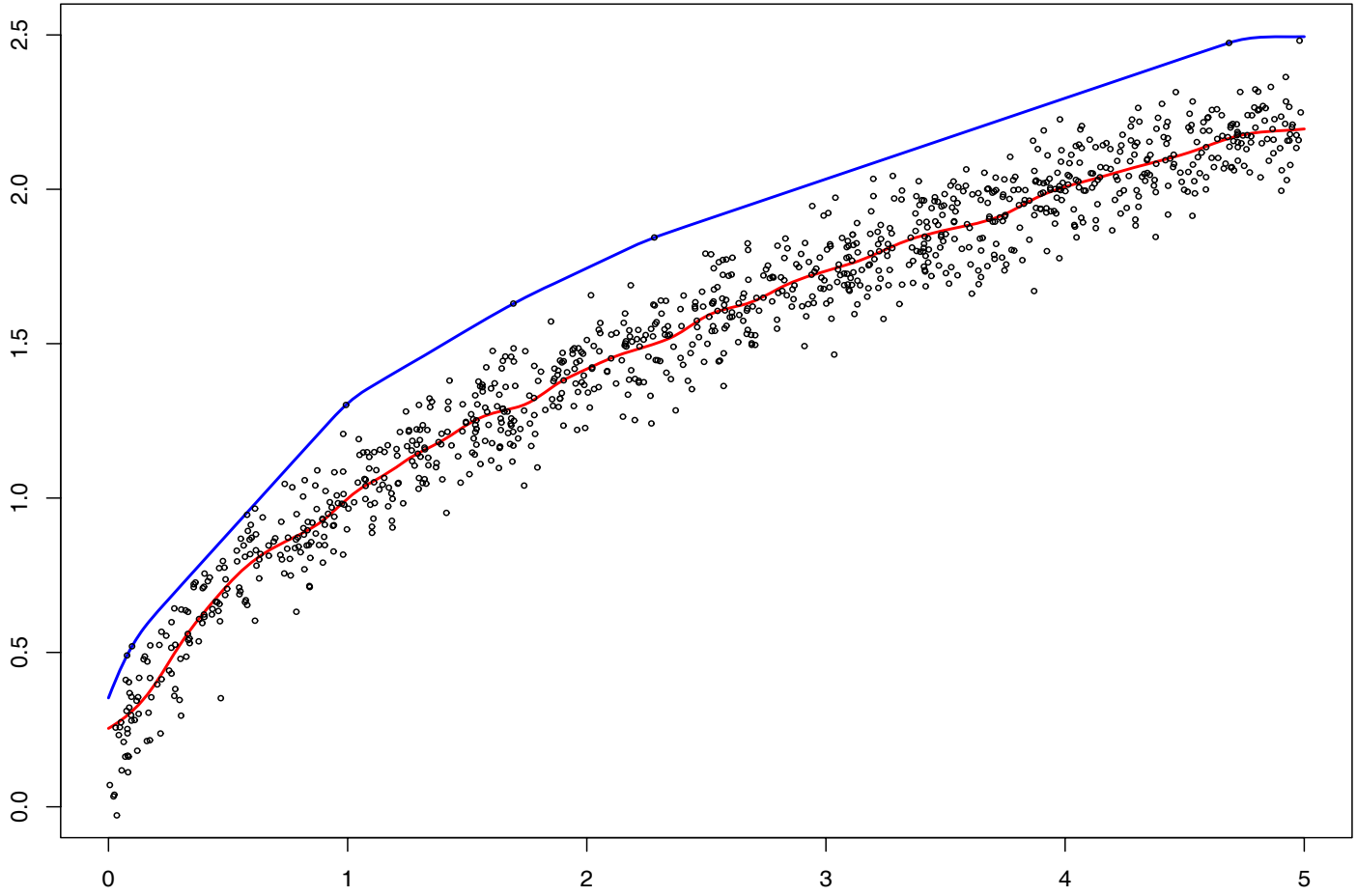
$g(\mathbf{x}) = \sin\left(\sqrt{x_1^2 + x_2^2}\right) / \sqrt{x_1^2 + x_2^2}$, $\partial \hat{g}^2(\mathbf{x}|p)/\partial x_1^2 \geq 0$, $\partial \hat{g}^2(\mathbf{x}|p)/\partial x_2^2 \geq 0$



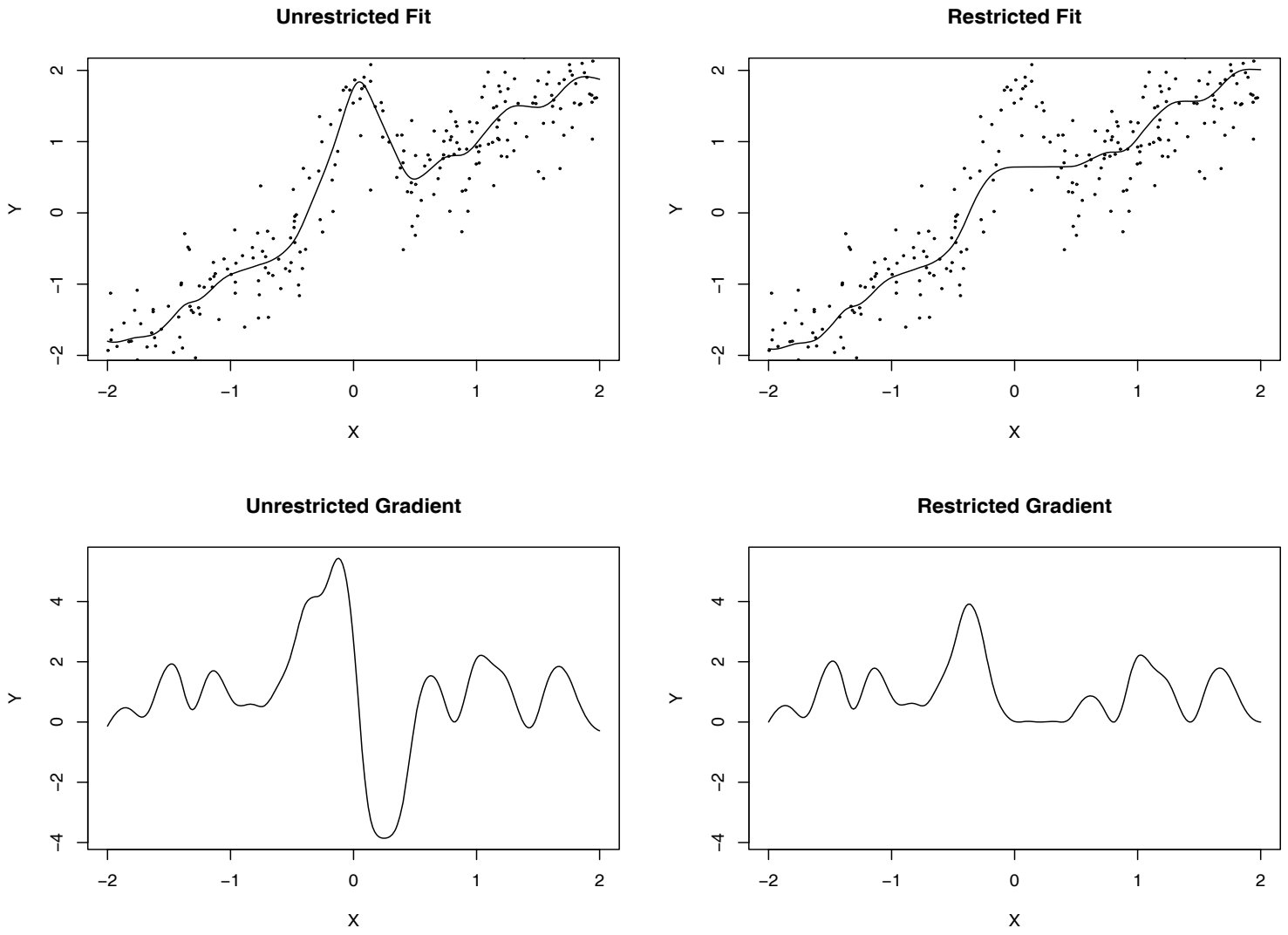
$g(\mathbf{x}) = \sin\left(\sqrt{x_1^2 + x_2^2}\right) / \sqrt{x_1^2 + x_2^2}$, $\partial \hat{g}^2(\mathbf{x}|p)/\partial x_1^2 \leq 0$, $\partial \hat{g}^2(\mathbf{x}|p)/\partial x_2^2 \leq 0$



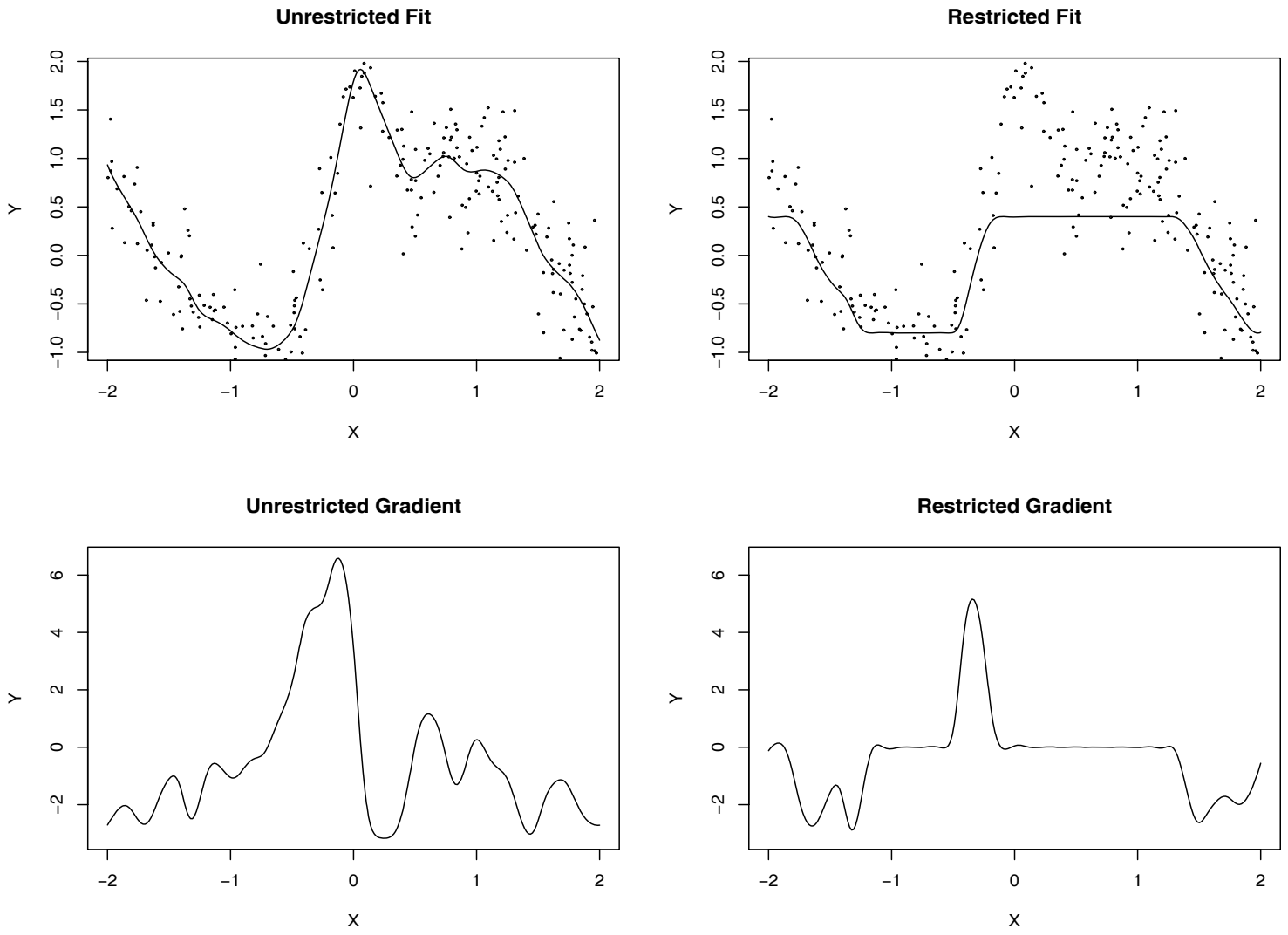
Production Frontier, $g(\mathbf{x}) = \sqrt{x}$, $y \leq \hat{g}(\mathbf{x}|p)$, $\hat{g}^1(\mathbf{x}|p) \geq 0$, $\hat{g}^2(\mathbf{x}|p) \leq 0$



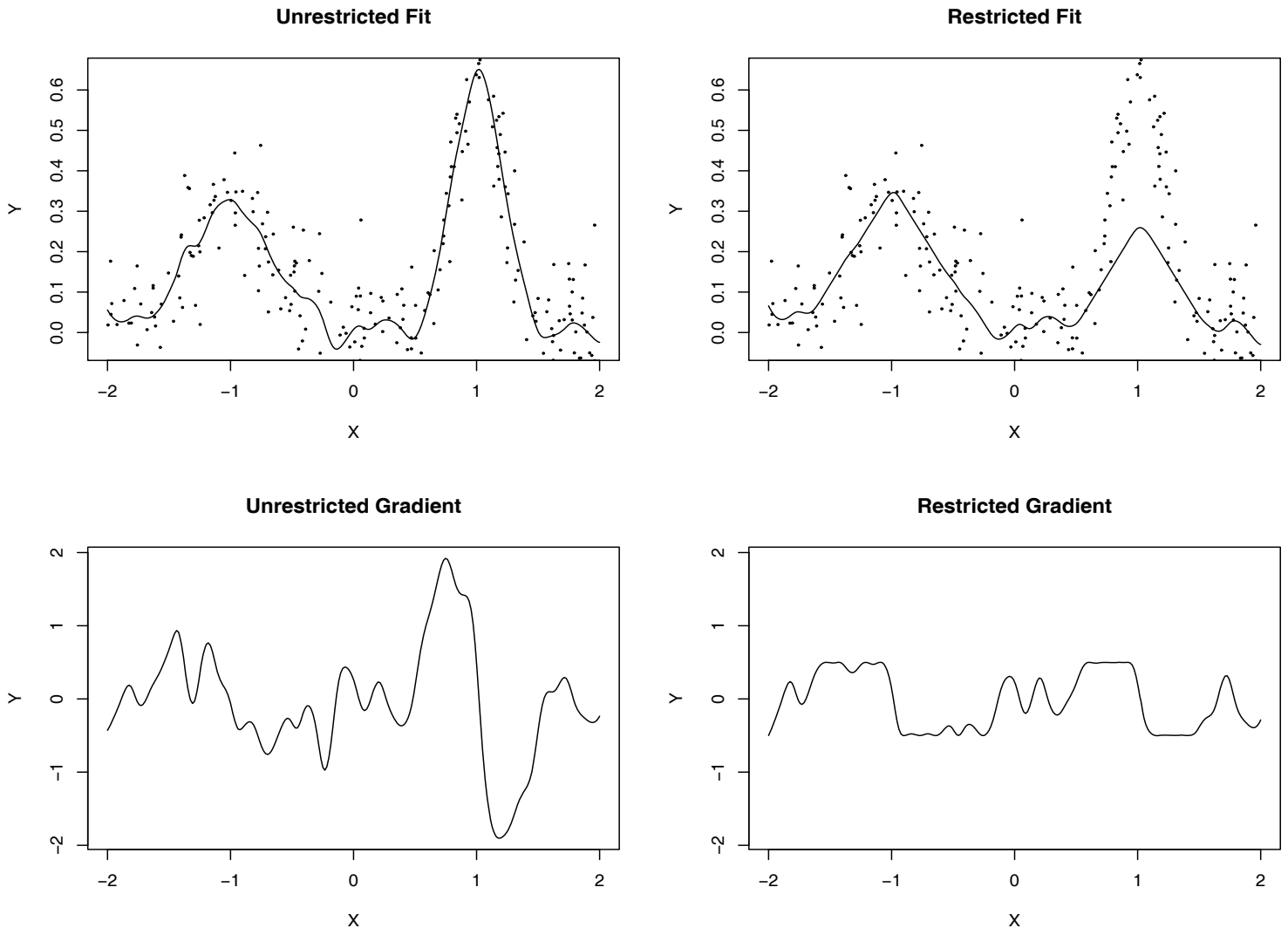
$g(\mathbf{x}) = x + 2e^{-16x^2}$, $\hat{g}^1(\mathbf{x}|p) \geq 0$



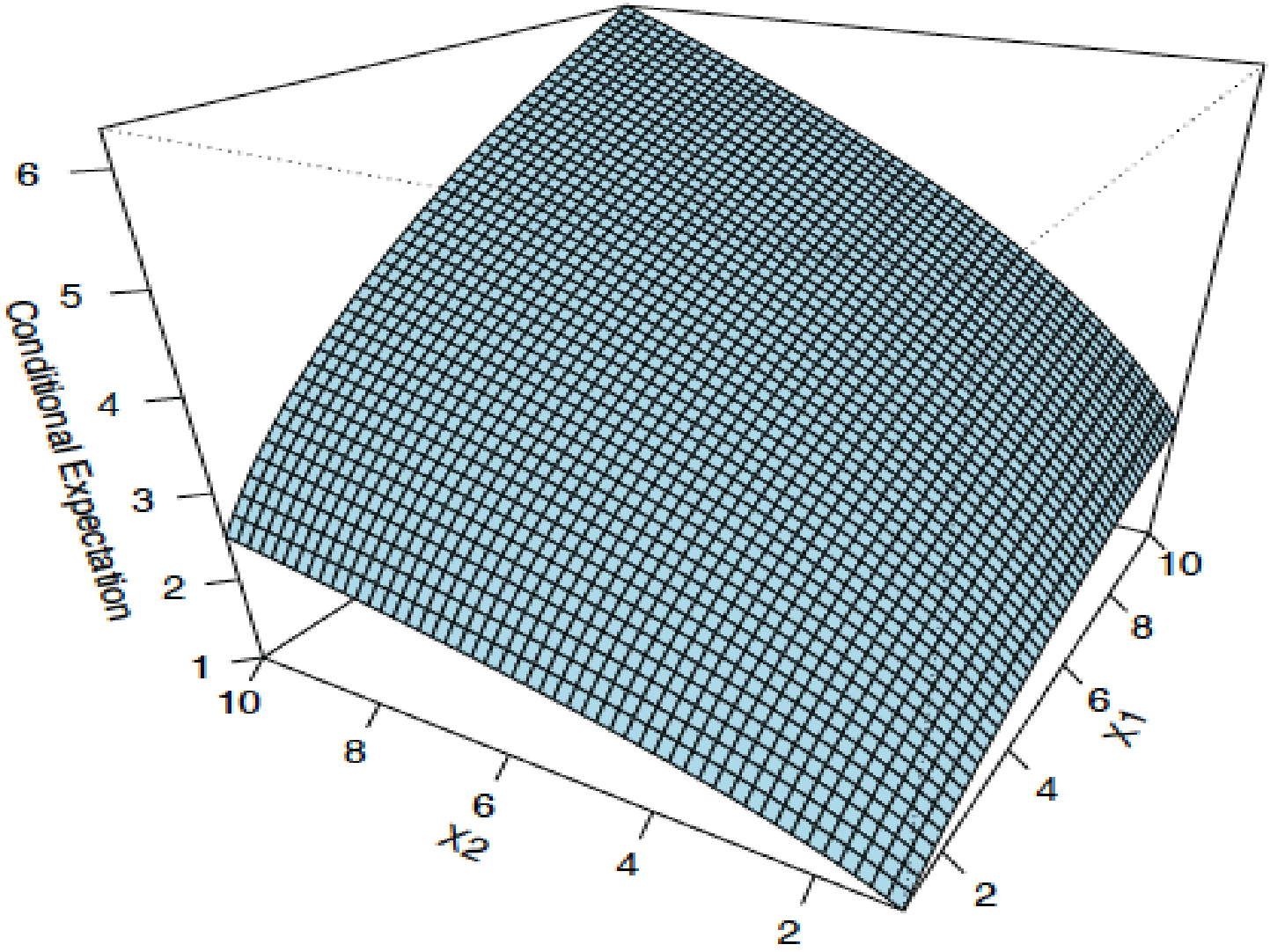
$g(\mathbf{x}) = \sin(2x) + 2e^{-16x^2}$, $-0.6 \leq \hat{g}(\mathbf{x}|p) \leq 0.6$



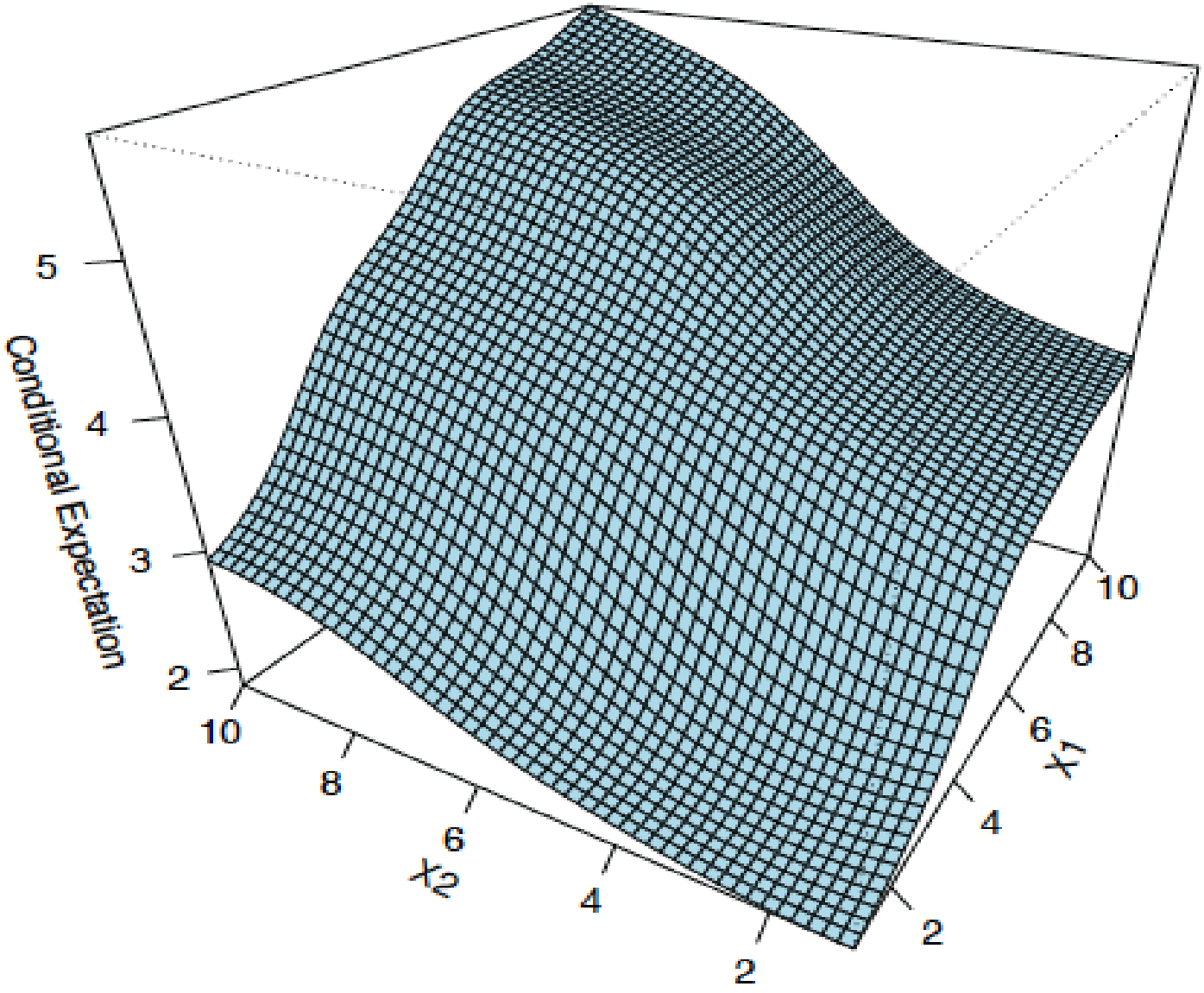
$g(\mathbf{x}) = 0.3e^{-4(x+1)^2} + 0.7e^{-16(x-1)^2}$, $-0.6 \leq \hat{g}^1(\mathbf{x}|p) \leq 0.6$



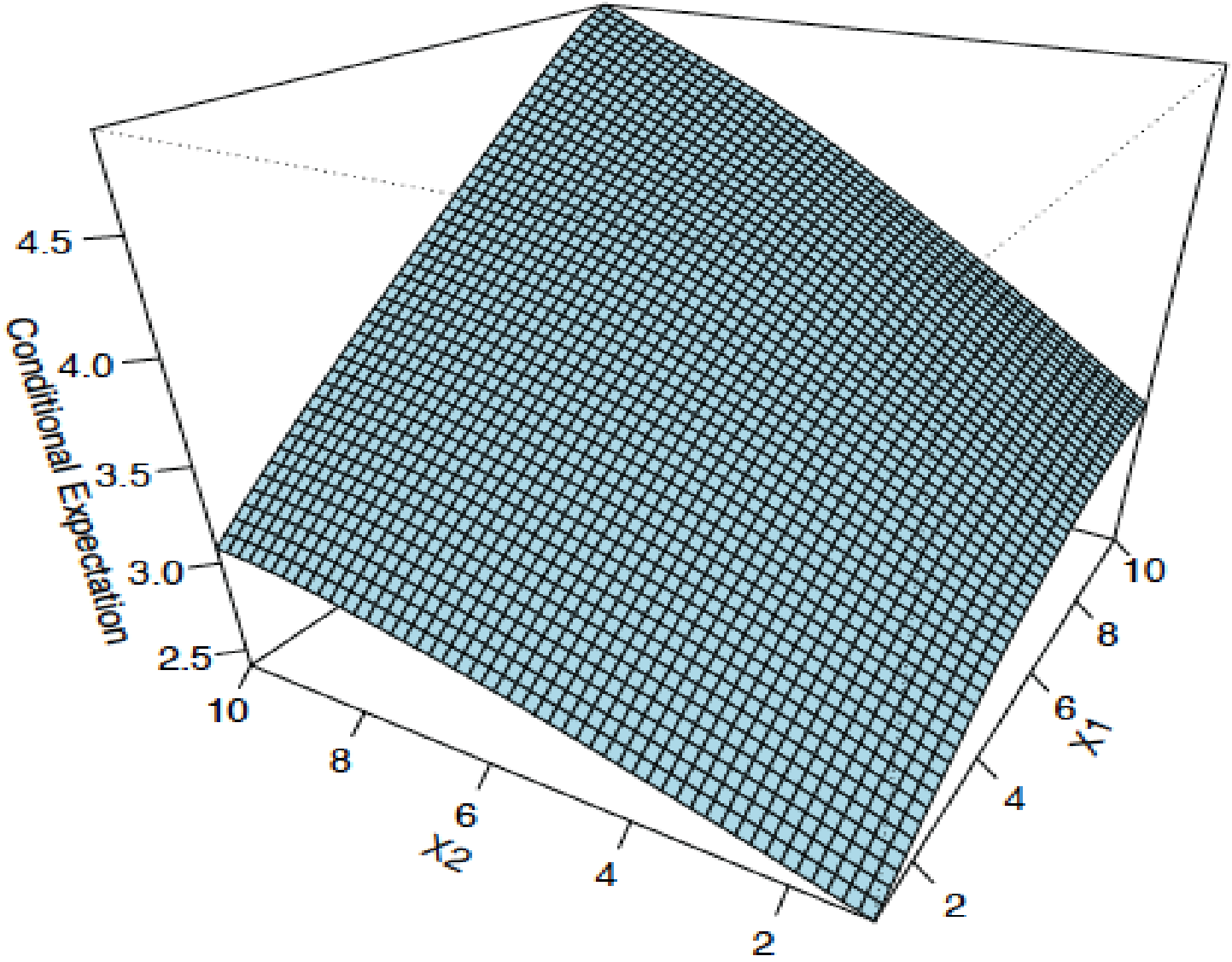
$g(\mathbf{x}) = (x_1 x_2)^{0.4}$



$g(\mathbf{x}) = (x_1 x_2)^{0.4}$, unrestricted estimate



$g(\mathbf{x}) = (x_1 x_2)^{0.4}$, $\partial \hat{g}^2(\mathbf{x}|p)/\partial x_1^2 \leq 0$, $\partial \hat{g}^2(\mathbf{x}|p)/\partial x_2^2 \leq 0$



Testing Constraint Validity

- We adopt a testing approach similar to that proposed by Hall, Huang, Gifford & Gijbels (2001) which is predicated on the objective function $D(\hat{p})$.
- This approach involves estimating the constrained regression function $\hat{g}(\mathbf{x}|p)$ based on the sample realizations $\{Y_i, \mathbf{X}_i\}$ and then rejecting H_0 if the observed value of $D(\hat{p})$ is too large.
- We use a resampling approach for generating the null distribution of $D(\hat{p})$ which involves generating resamples for y drawn from the constrained model via *iid* residual resampling (i.e., conditional on the sample $\{\mathbf{X}_i\}$), which we denote $\{Y_i^*, \mathbf{X}_i\}$.
- These resamples are generated under H_0, hence we recompute $\hat{g}(\mathbf{x}|p)$ for the bootstrap sample $\{Y_i^*, X_i\}$ which we denote $\hat{g}(\mathbf{x}|p^*)$ which then yields $D(p^*)$.

Testing Constraint Validity

- We then repeat this process B times.
- Finally, we compute the empirical P value, P_B, which is simply the proportion of the B bootstrap resamples $D(p^*)$ that exceed $D(\hat{p})$, i.e.,

$$P_B = 1 - \hat{F}(D(\hat{p})) = \frac{1}{B}\sum_{j=1}^{B} I(D(p^*) > D(\hat{p})),$$

  where $I(\cdot)$ is the indicator function and $\hat{F}(D(\hat{p}))$ is the EDF of the bootstrap statistics.
- Then one rejects the null hypothesis if P_B is less than α, the level of the test.

Asymptotic Property of $D(\hat{p})$

Let $\psi_i(\mathbf{x}) = A_i^{\mathcal{D}}(\mathbf{x})Y_i, i = 1, \ldots, n$. Recall that our minimization problem is

$$\min_{p_1,\ldots,p_n} \Sigma_{i=1}^n (n^{-1} - p_i)^2, \quad \text{s.t.} \quad \Sigma_{i=1}^n p_i = 1, \Sigma_{i=1}^n p_i \psi_i(\mathbf{x}) \geq 0, \forall \mathbf{x}.$$

In practice, this minimization is carried out by taking a fine grid $(x_1, ..., x_N)$, where N is large, and solving

$$\min_{p_1,\ldots,p_n} \Sigma_{i=1}^n (n^{-1} - p_i)^2,$$
$$\text{s.t.} \quad \Sigma_{i=1}^n p_i = 1, \Sigma_{i=1}^n p_i \psi_i(\mathbf{x}_j) \geq 0, 1 \leq j \leq N.$$

Asymptotic Property of $D(\hat{p})$

Assumption (Assumption 3.2)

(i) $N \to \infty$ as $n \to \infty$ and $N = O(n)$.

(ii) Let $d_N = \inf_{1 \le j_1, j_2 \le N} |x_{j_1} - x_{j_2}|$ be the minimum distance between grid points. We require $d_N \to 0$ and $h^{-1} d_N \to \infty$.

Theorem (Theorem 3.3)

Suppose that assumptions 3.1(i)-(iv) and 3.2(i)-(ii) hold. Then

$$\frac{n^2 \sigma_K^2}{h^{2|\mathbf{d}|+r} \left(\sum_{j=1}^{M} g^{\mathcal{D}}(\mathbf{x}_j^*) \right)^2} D(\hat{p}) \sim \chi^2(n)$$

as $n \to \infty$, where $\sigma_K^2 = \sigma^2 \int [K^{(\mathbf{d})}(y)]^2 dy$, and $\{\mathbf{x}_1^*, ..., \mathbf{x}_M^*\} \subset \{\mathbf{x}_1, ..., \mathbf{x}_N\}$ are the slack points defined in Appendix A.3.

Consistency of Hypothesis Test

Similar to Carroll, Delaigle and Hall (2011), we can have

(i) If the true function g satisfies the shape constraint, then as $n \to \infty$,

$$P\{D(\hat{p}) \leq n\epsilon\} \to 1 \quad \forall \epsilon > 0.$$

(ii) If the true function g does not satisfy the shape constraint on $\mathcal{J}$, then

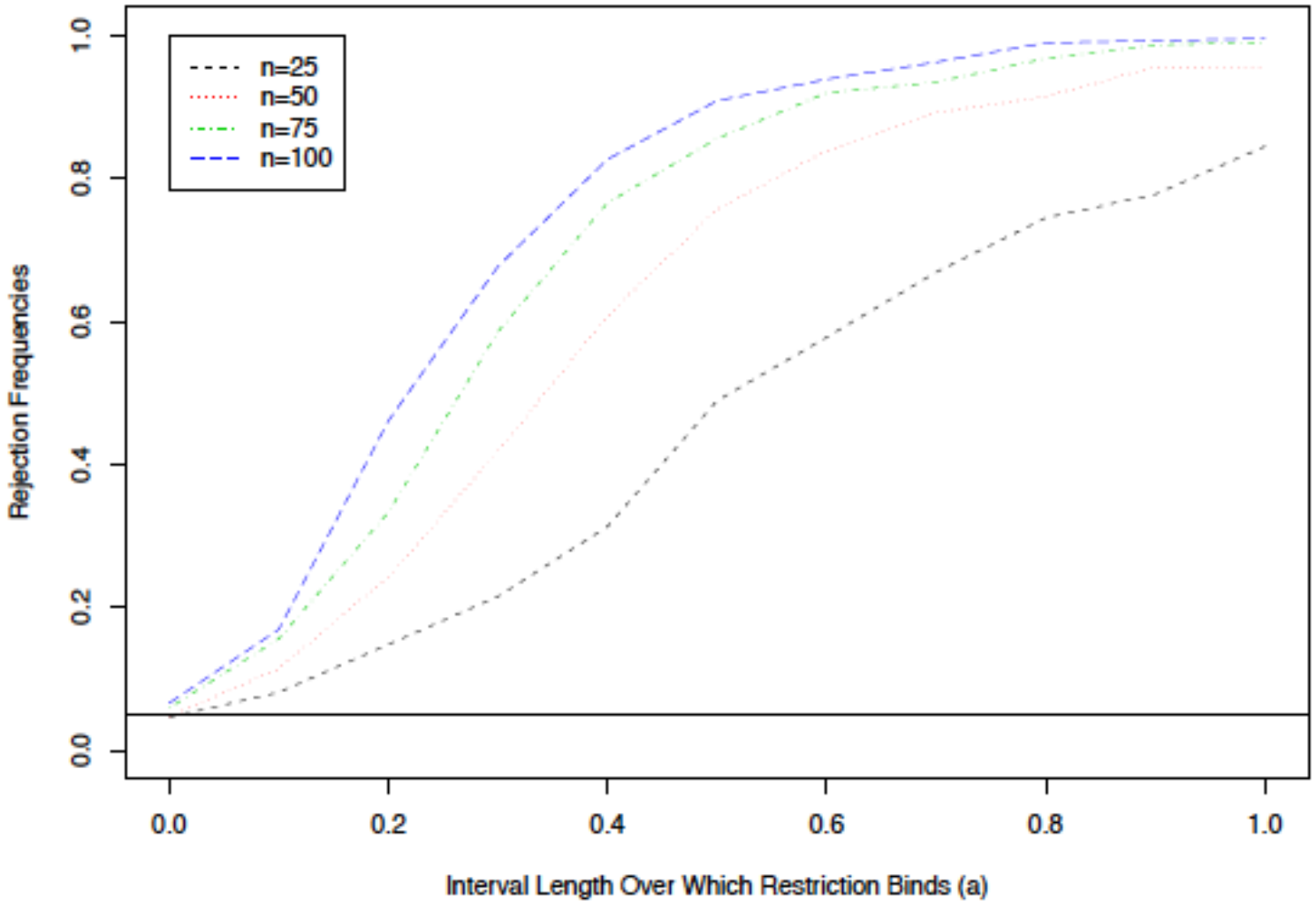
$$\lim_{\epsilon \to 0} \liminf_{n \to \infty} P\{D(\hat{p}) \geq n\epsilon\} = 1.$$

Simulations: Testing Inequality Restrictions

- $H_0 : g(\mathbf{x}) \geq 0$ versus $H_1 : g(\mathbf{x}) < 0$.
- We consider the following simple linear DGP:

$$Y_i = g(X_i) + \varepsilon_i = X_i + \varepsilon_i,$$

  where X_i, is uniform $[-a, 4-a]$, a is a parameter determining whether the constraint is binding $(a > 0)$ or not $(a = 0)$ and the length of interval over which the constraint binds, and $\varepsilon \sim N(0, 1/4)$
- When $a = 0$, we can assess the test's size.
- When $a > 0$, we can assess the test's power.
- We present power curves for this example to illustrate the size-power comparison for $\alpha = 0.05$.


n=25
n=50
n=75
n=100
Rejection Frequencies
0.0
0.2
0.4
0.6
0.8
1.0
Interval Length Over Which Restriction Binds (a)

Simulations: Testing Equality Restrictions

- We consider the following simple linear DGP:

$$Y_i = g(X_{i1}, X_{i2}) + \varepsilon_i = 1 + X_{i1}^2 + X_{i2} + \varepsilon_i,$$

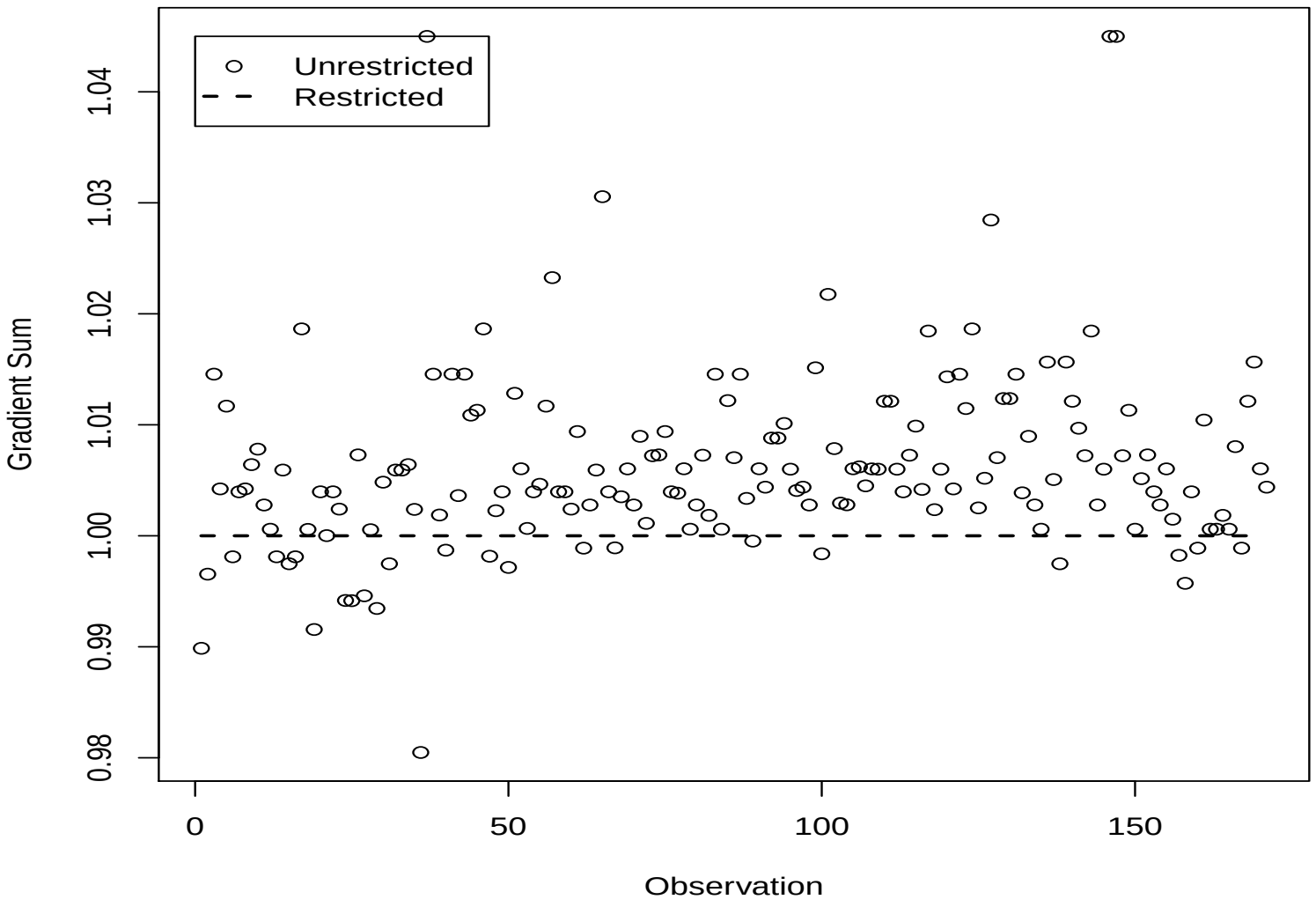
  where $X_i j$, j=1,2 are uniform $[-2, 2]$, and $\varepsilon \sim N(0, 1/2)$.
- $H_0 : g(\mathbf{x})$ is a linear function x_1^2 and x_2. (assess the test's size)
- $H_0 : g(\mathbf{x})$ is a linear function x_1 and x_2. (assess the test's power)

n	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
		Size	
25	0.100	0.049	0.010
50	0.074	0.043	0.011
75	0.086	0.034	0.008
100	0.069	0.031	0.006
200	0.093	0.044	0.007
		Power	
25	0.391	0.246	0.112
50	0.820	0.665	0.356
75	0.887	0.802	0.590
100	0.923	0.849	0.669
200	0.987	0.970	0.903

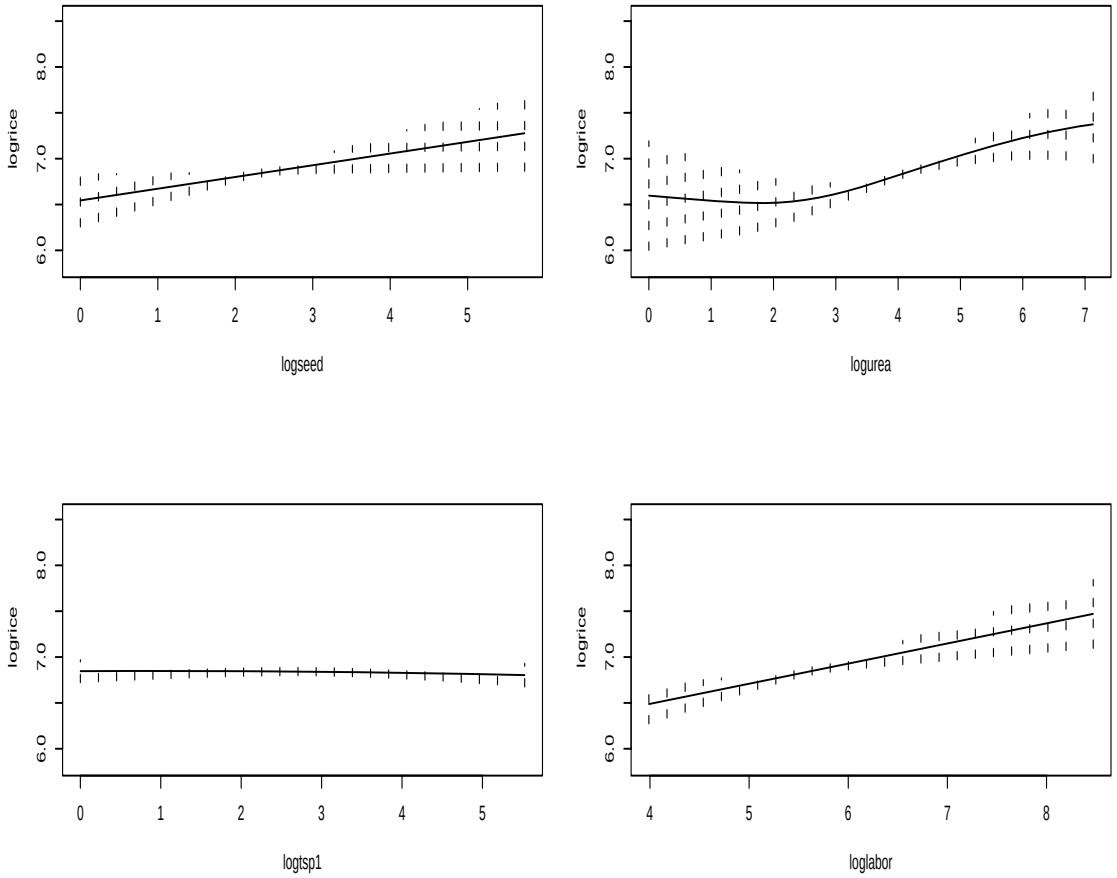
- We consider a production dataset that has been studied by Horrace & Schmidt (2000) who analyzed technical efficiency for Indonesian rice farms.
- We examine the issue of *returns to scale*, focusing on one growing season's worth of data for the year 1977, acknowledged to be a particularly wet season.
- Farmers were selected from six villages of the production area of the Cimanuk River Basin in West Java, and there were 171 farms in total.
- Output is measured as kilograms of rice produced, and inputs included seed (kg), urea (kg), trisodium phosphate (TSP) (kg), labour (hours), and land (hectares).
- We use log transformations throughout.

Returns to Scale

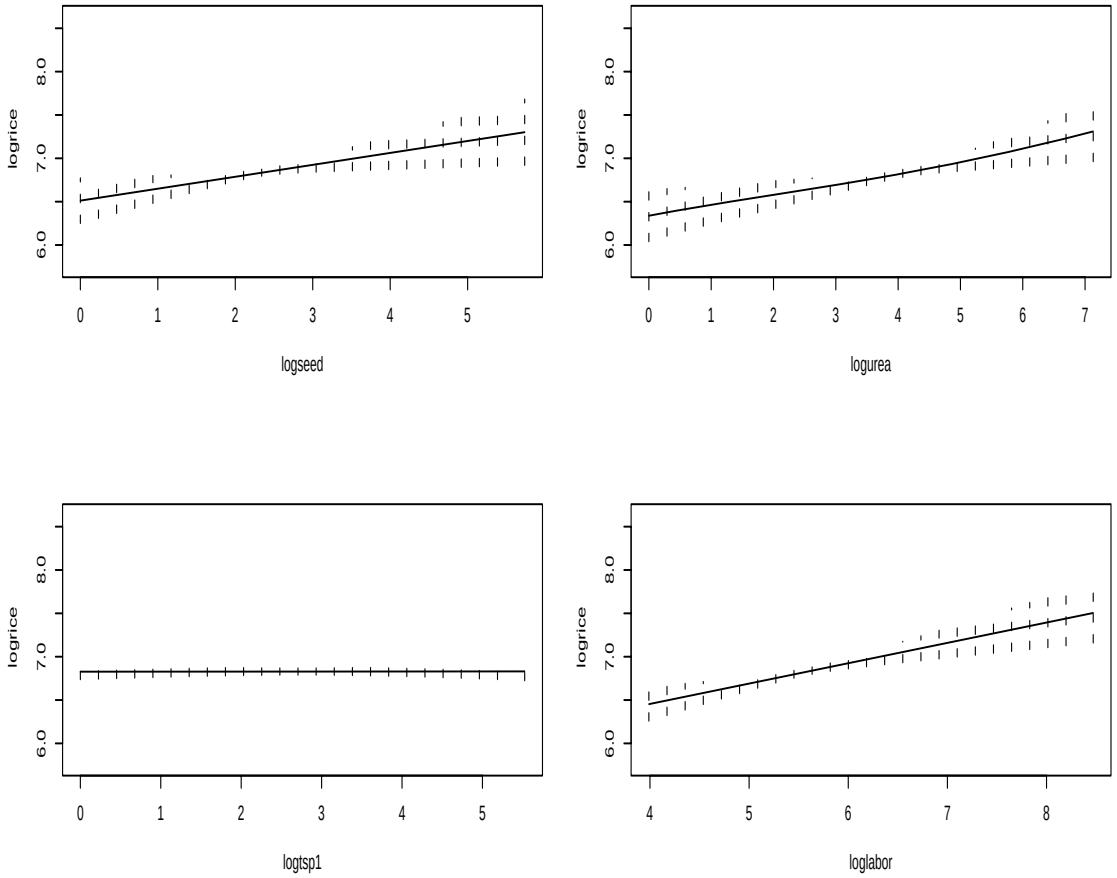
- Of interest here is whether or not the production technology exhibits constant returns to scale (i.e., whether or not the sum of the first order partial derivatives equals one).
- Returns to scale measure the proportional increase in output for an equal increase in all inputs. Constant returns to scale implies that output increases by exactly the amount that all the inputs are increased; if all the inputs are doubled, then output doubles.
- Given the primitive nature of the production of rice, one expects the existence of constant returns to scale to be present in the underlying technology.
- Returns to scale can provide insights into appropriate rural development strategies. If several farms possessed decreasing returns to scale then farm operations could be scaled back resulting in lower costs.


Unrestricted
Restricted
Gradient Sum
0.98
0.99
1.00
1.01
1.02
1.03
1.04
0
50
100
150
Observation

Unrestricted



Restricted


logrice
6.0
7.0
8.0
0
1
2
3
4
5
logseed
logrice
6.0
7.0
8.0
0
1
2
3
4
5
6
7
logurea
logrice
6.0
7.0
8.0
0
1
2
3
4
5
logtsp1
logrice
6.0
7.0
8.0
4
5
6
7
8
loglabor

- In order to test whether the restriction is valid we apply the proposed test.
- We conducted $B = 99$ bootstrap replications and test the null that the technology exhibits constant returns to scale.
- The empirical P value is $P_B = 0.131$, hence we fail to reject the null at all conventional levels.
- We are encouraged by this fully nonparametric application particularly as it involves a fairly large number of regressors (five) and a fairly small number of observations ($n = 171$).

- We present a framework for imposing and testing the validity of arbitrary constraints on the sth partial derivatives of a nonparametric kernel regression function.
- The proposed approach nests special cases such as imposing monotonicity, concavity (convexity) and so forth while delivering a seamless framework for general restricted nonparametric kernel estimation and inference.
- Illustrative simulated examples are presented, finite-sample performance of the proposed test is examined via Monte Carlo simulations, and an illustrative application is undertaken.
- Available as a contributed R package `np` maintained by Jeffrey Racine.